



RESTAURANT SALES ANALYSIS

USING SQL



INTRODUCTION



➤ OBJECTIVES

Analyze Restaurant Sales Data Using SQL

➤ GOALS

Understand sales trends

Identify top-selling items

Analyze revenue and customer behavior

QUESTION 1

--1. View the menu_items table and write a query to find the number of items on the menu

```
SELECT COUNT(*) AS number_of_items  
FROM menu_items
```

	number_of_items bigint 
1	32

QUESTION 2

--2. What are the least and most expensive items on the menu?

--most expensive item

```
SELECT item_name, price
FROM menu_items
ORDER BY price DESC
LIMIT 1
```

	item_name character varying (45) 🔒	price numeric (5,2) 🔒
1	Shrimp Scampi	19.95

--least expensive item

```
SELECT item_name, price
FROM menu_items
ORDER BY price
LIMIT 1
```

	item_name character varying (45) 🔒	price numeric (5,2) 🔒
1	Edamame	5.00

QUESTION 3

--3. How many Italian dishes are on the menu? What are the least and most expensive Italian dishes on the menu?

```
SELECT COUNT(*) AS number_of_italian_dishes
FROM menu_items
WHERE category = 'Italian'
```

	number_of_italian_dishes bigint
1	9

```
SELECT category,item_name AS most_italian_dish, price
FROM menu_items
WHERE category = 'Italian'
ORDER BY price DESC
LIMIT 1
```

	category character varying (45)	most_italian_dish character varying (45)	price numeric (5,2)
1	Italian	Shrimp Scampi	19.95

```
SELECT category,item_name AS least_italian_dish, price
FROM menu_items
WHERE category = 'Italian'
ORDER BY price
LIMIT 1
```

	category character varying (45)	least_italian_dish character varying (45)	price numeric (5,2)
1	Italian	Spaghetti	14.50

QUESTION 4

--4. How many dishes are in each category? What is the average dish price within each category?

```
SELECT category, COUNT(*) AS dishes_count  
FROM menu_items  
GROUP BY category
```

	category character varying (45) 🔒	dishes_count bigint 🔒
1	American	6
2	Mexican	9
3	Asian	8
4	Italian	9

```
SELECT category, ROUND(AVG(price),2) AS average_price  
FROM menu_items  
GROUP BY category
```

	category character varying (45) 🔒	average_price numeric 🔒
1	American	10.07
2	Mexican	11.80
3	Asian	13.48
4	Italian	16.75

QUESTION 5

--5. View the order_details table. What is the date range of the table?

```
SELECT * FROM order_details
```

```
SELECT  
    MIN(order_date) AS earliest_date,  
    MAX(order_date) AS latest_date  
FROM  
    order_details
```

	earliest_date date	latest_date date
1	2023-01-01	2023-03-31

QUESTION 6

--6. How many orders were made within this date range? How many items were ordered within this date range?

```
SELECT COUNT (order_id) AS Number_of_orders, COUNT(item_id) AS  
items_ordered
```

```
FROM order_details
```

```
WHERE order_date BETWEEN '2023-01-01' AND '2023-03-31'
```

	number_of_orders  bigint	items_ordered  bigint
1	12234	12097

QUESTION 7

--7. Which orders had the most number of items?

```
SELECT order_id, COUNT(item_id) AS number_of_items
FROM order_details
GROUP BY order_id
ORDER BY COUNT(item_id) DESC
LIMIT 7
```

	order_id smallint	number_of_items bigint
1	4305	14
2	1957	14
3	2675	14
4	3473	14
5	440	14
6	443	14
7	330	14

QUESTION 8

--8. How many orders had more than 12 items?

```
SELECT COUNT (*) AS orders_more_than_12
FROM (SELECT order_id, COUNT(item_id) AS number_of_items
FROM order_details
GROUP BY order_id
HAVING COUNT(item_id) > 12)
```

	orders_more_than_12 
1	20



QUESTION 9

--9. Combine the menu_items and order_details tables into a single table

```
SELECT * FROM order_details od  
JOIN menu_items mi ON od.item_id = mi.menu_item_id
```

QUESTION 10

10. What were the least and most ordered items? What categories were they in?

--most

```
SELECT mi.category, mi.item_name, COUNT(DISTINCT order_id) AS Ordered_items
FROM order_details od
JOIN menu_items mi ON od.item_id = mi.menu_item_id
GROUP BY mi.category, mi.item_name
ORDER BY ordered_items DESC
LIMIT 1
```

	category character varying (45) 🔒	item_name character varying (45) 🔒	ordered_items bigint 🔒
1	Asian	Edamame	596

--least

```
SELECT mi.category, mi.item_name, COUNT(DISTINCT order_id) AS Ordered_items
FROM order_details od
JOIN menu_items mi ON od.item_id = mi.menu_item_id
GROUP BY mi.category, mi.item_name
ORDER BY ordered_items
LIMIT 1
```

	category character varying (45) 🔒	item_name character varying (45) 🔒	ordered_items bigint 🔒
1	Mexican	Chicken Tacos	123

QUESTION 11

11.What were the top 5 orders that spent the most money?

```
SELECT od.order_id, SUM(price) AS most_spent
FROM order_details od JOIN menu_items mi ON mi.menu_item_id = od.item_id
GROUP BY od.order_id
ORDER BY SUM(price) DESC
LIMIT 5
```

	order_id smallint 🔒	most_spent numeric 🔒
1	440	192.15
2	2075	191.05
3	1957	190.10
4	330	189.70
5	2675	185.10

QUESTION 12

--12. View the details of the highest spend order. Which specific items were purchased?

```
SELECT category, item_name, od.order_id, SUM(price) AS most_spent
FROM order_details od JOIN menu_items mi ON mi.menu_item_id = od.item_id
GROUP BY category, item_name, od.order_id
ORDER BY SUM(price) DESC
LIMIT 1
```

	category character varying (45) 🔒	item_name character varying (45) 🔒	order_id smallint 🔒	most_spent numeric 🔒
1	Asian	Korean Beef Bowl	1639	53.85

```
SELECT category, item_name, SUM(price) AS most_spent
FROM order_details od JOIN menu_items mi ON mi.menu_item_id = od.item_id
GROUP BY category, item_name
ORDER BY SUM(price) DESC
LIMIT 5
```

	category character varying (45) 🔒	item_name character varying (45) 🔒	most_spent numeric 🔒
1	Asian	Korean Beef Bowl	10554.60
2	Italian	Spaghetti & Meatballs	8436.50
3	Asian	Tofu Pad Thai	8149.00
4	American	Cheeseburger	8132.85
5	American	Hamburger	8054.90

QUESTION 13

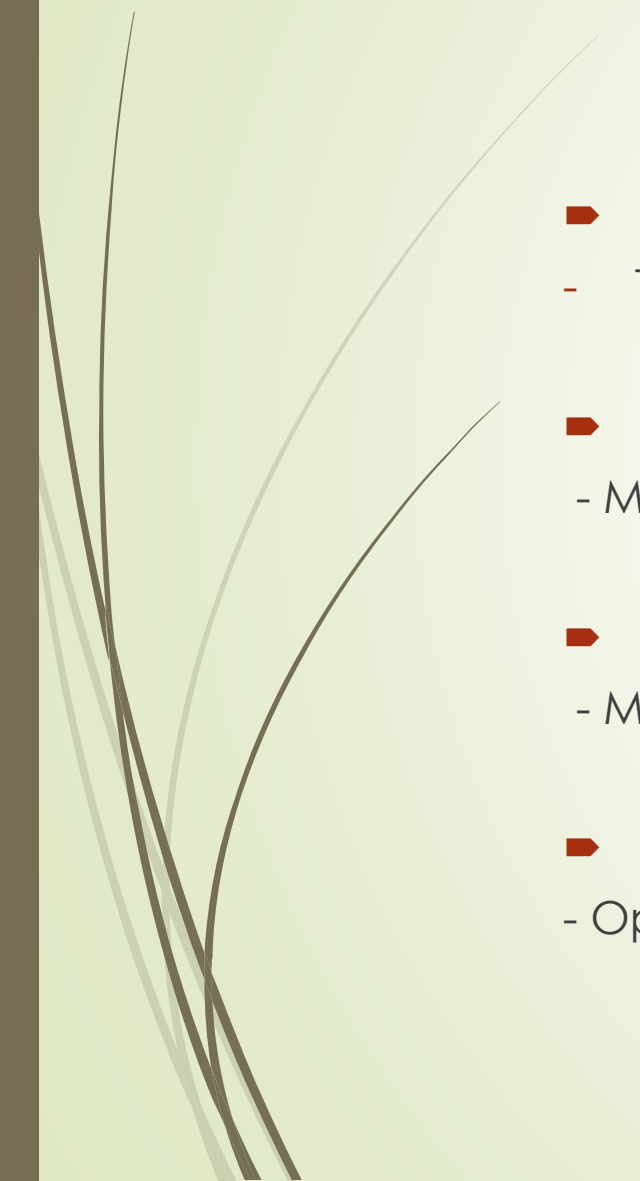
--13. View the details of the top 5 highest spend orders

```
SELECT category, item_name, od.order_id, SUM(price) AS most_spent
FROM order_details od JOIN menu_items mi ON mi.menu_item_id = od.item_id
GROUP BY category, item_name, od.order_id
ORDER BY SUM(price) DESC
LIMIT 5
```

	category character varying (45) 🔒	item_name character varying (45) 🔒	order_id smallint 🔒	most_spent numeric 🔒
1	Italian	Meat Lasagna	2364	53.85
2	Asian	Korean Beef Bowl	1639	53.85
3	Italian	Chicken Parmesan	740	53.85
4	Italian	Eggplant Parmesan	3940	50.85
5	Italian	Eggplant Parmesan	4314	50.85



REPORTS & INSIGHTS



- ▶ • Sales Summary Report:
 - Total revenue, number of orders, average order value
- ▶ • Customer Insights:
 - Most frequent customers, high-value customers
- ▶ • Popular Items:
 - Most ordered dishes, seasonal trends
- ▶ • Recommendations:
 - Optimize menu pricing, target high-value customers